

Android/Mobile Cryptography Benchmark Analysis Report

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Scope

This report summarizes the benchmark analysis performed in the notebook for the mobile cryptography experiment. The analysis uses the raw round-level CSV files in unplugged_run/d1 and unplugged_run/d2 as the mobile datasets, and the CSV files in the time folder as the PC/computer baseline.

The raw mobile files contain the expected benchmark schema:

- size_bytes
- round_index
- enc_ns
- dec_ns
- enc_mem_bytes
- dec_mem_bytes
- energy_mWh
- method

The notebook converts time to milliseconds, energy to Joules, and size to KB/MB, then computes aggregated summaries and rankings.

Data Quality Summary

The raw mobile benchmark dataset contains 900 rows in total after combining the raw round-level files from both devices.

Main quality checks:

- Missing values: none detected.
- Duplicated rows: none detected.
- Invalid numeric values: none detected.
- Expected rounds per size/algorithm/device combination: 5.
- IQR-based outliers: 300 rows flagged across the dataset.

The outlier flag is kept for interpretation and boxplot analysis; the original observations are not removed.

Key Derived Metrics

The notebook computes the following derived metrics:

- $\text{enc_ms} = \text{enc_ns} / 1,000,000$
- $\text{dec_ms} = \text{dec_ns} / 1,000,000$
- $\text{total_time_ms} = \text{enc_ms} + \text{dec_ms}$
- $\text{energy_J} = \text{energy_mWh} * 3.6$
- $\text{size_kb} = \text{size_bytes} / 1024$
- $\text{size_mb} = \text{size_bytes} / 1024^2$
- avg_mem_bytes = average of encryption and decryption memory usage
- $\text{total_mem_bytes} = \text{enc_mem_bytes} + \text{dec_mem_bytes}$

Main Mobile Results

Fastest Algorithm per Device

Device	Best algorithm	Mean total time (ms)	Mean energy (J)
D1	AES-GCM	6.3307	0.000430
D2	AES-GCM	6.2992	0.000486

AES-GCM is the fastest algorithm on both devices, and the result is consistent across the two mobile platforms.

Lowest-Energy Algorithm per Device

Device	Best algorithm	Mean total time (ms)	Mean energy (J)
D1	ElGamal	38.5726	0.000174
D2	ElGamal	38.4249	0.000212

ElGamal minimizes energy on both devices, but it does so at a much higher runtime cost than AES-GCM.

Best Algorithm by Input Size

Input size (bytes)	Best algorithm	Mean total time (ms)	Mean energy (J)
1024	ChaCha20	2.8039	0.000185
2048	ChaCha20	3.0398	0.000114
4096	AES-GCM	3.1630	0.000123
8192	AES-GCM	4.0590	0.000175
16384	AES-GCM	4.5586	0.000231
32768	AES-GCM	5.2747	0.000319

Input size (bytes)	Best algorithm	Mean total time (ms)	Mean energy (J)
65536	AES-GCM	6.5562	0.000514
131072	AES-GCM	10.0832	0.000871
262144	AES-GCM	17.2153	0.001586

ChaCha20 is the best at the smallest sizes, but AES-GCM becomes the dominant choice for medium and large inputs.

Trade-Off Ranking

The notebook computes a normalized score using:

- $\text{time_norm} = \text{min-max normalized total_time_ms}$
- $\text{energy_norm} = \text{min-max normalized energy_J}$
- $\text{score} = \alpha * \text{time_norm} + (1 - \alpha) * \text{energy_norm}$

with $\alpha = 0.5$ by default.

Composite Ranking

Rank	Algorithm	Mean score	Median score	Score std	Mean total time (ms)	Mean energy (J)
1	AES-GCM	0.0207	0.0080	0.0272	6.3150	0.000458
2	ElGamal	0.0242	0.0176	0.0138	38.4988	0.000193
3	RSA-Hybrid	0.0249	0.0183	0.0137	37.9185	0.000213
4	ASCON	0.0275	0.0094	0.0358	12.8408	0.000521
5	Xoodyak	0.0341	0.0102	0.0481	16.2028	0.000613
6	ChaCha20	0.0386	0.0103	0.0570	9.6171	0.000768
7	GIFT-COFB	0.0613	0.0229	0.0780	52.1101	0.000766
8	AES	0.0774	0.0275	0.1048	12.1958	0.001488
9	Grain-128AEAD	0.1234	0.0356	0.1720	90.6167	0.001563
10	Elephant	0.2293	0.0731	0.3109	225.9220	0.002201

Sensitivity to alpha

alpha	Best algorithm
0.25	ElGamal
0.50	AES-GCM
0.75	AES-GCM

AES-GCM is the best overall choice for the default balanced weighting and remains the best when time is weighted more heavily.

PC / Computer Baseline Comparison

The time folder corresponds to the PC/computer benchmark run. It is not the same as the mobile benchmark data, so it is treated as a separate baseline and compared only for algorithms that appear in both datasets.

PC vs Mobile Summary

The comparison is plotted with log scales, so the PC results remain visible even where the absolute values differ strongly from the mobile runs.

- AES-GCM is the fastest algorithm on both platforms.
- The mobile runtime penalty is smallest for AES, at about 1.84× slower than PC.
- The mobile runtime penalty is largest for ElGamal, at about 16.43× slower than PC.
- RSA-Hybrid is also much slower on mobile, at about 12.56× slower than PC.
- ChaCha20 stays relatively close to the PC baseline, with a mobile slowdown of about 3.18×.
- Energy values are much smaller on mobile because the notebook converts the benchmark energy readings into Joules and the devices were measured under different execution conditions.

PC-only Ranking

Rank	Algorithm	Mean time (ms)	Mean energy (J)
1	AES-GCM	8.3932	0.0563
2	ChaCha20	16.1282	0.0922
3	AES	18.3687	0.1005
4	RSA-Hybrid	18.5917	0.1063
5	ElGamal	18.9334	0.1063

The PC-only summary makes the computer baseline explicit and shows that AES-GCM remains the best all-round choice even on the desktop run, while the other algorithms cluster at higher runtime and energy values.

Variability and Stability

The notebook also generates round-level boxplots for total time.

Main takeaway:

- AES-GCM and ChaCha20 show comparatively tighter distributions.
- Larger and more complex algorithms exhibit wider spread and more variability across rounds.
- The IQR-based outlier flags support the visual interpretation that some algorithms are more stable than others under repeated execution.

Figures Generated

The following plots were created and saved as PNG files:

- plots/time_trends_by_algorithm_device.png
- plots/energy_memory_trends.png
- plots/algorithm_comparison_by_device.png
- plots/device_comparison_by_algorithm.png
- plots/round_level_variability_total_time.png
- plots/pc_vs_mobile_time.png
- plots/pc_vs_mobile_energy.png
- plots/pc_vs_mobile_ratios.png
- plots/pc_algorithm_summary.png
- plots/device_d1_summary.png
- plots/device_d2_summary.png

Conclusions

1. AES-GCM is the best overall algorithm for the mobile benchmark when balancing time and energy with $\alpha = 0.5$.
2. ElGamal is the lowest-energy algorithm on both devices, but it is far slower and therefore not the best balanced choice.
3. ChaCha20 is best for the smallest input sizes, while AES-GCM dominates most medium and large input sizes.
4. The PC/computer baseline is faster than the mobile devices for the shared algorithms, and the log-scale PC comparison plots make that difference visible in the figures.
5. The per-device figures make it easy to include a separate summary page for each device in the final report.

Exported Artifacts

The notebook exports the following files under `benchmark_analysis_outputs`:

- tables/aggregated_results.csv
- tables/summary_rankings.csv
- tables/tradeoff_ranking.csv
- tables/pc_algorithm_ranking.csv
- tables/pc_vs_mobile_comparison.csv
- all PNG plots listed above

Final Note

If you need this report adapted into a more formal thesis style, the same content can be rewritten into a short academic discussion section with references to the saved figures and tables.